

Your grade: 88%

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COACH

Almost there! Let's review your answers and then try again.

Your work showed a strong understanding of overfitting prevention techniques, LSTM advantages, dropout benefits, Markov processes, and the application of Bellman equations. However, there are a couple of areas where further review could enhance your understanding. Before retrying this assignment, we recommend focusing on these key areas:

- **Deep Neural Networks with RL:** In Question 4, your understanding of why actions are selected randomly based on their probabilities in a neural network policy was not accurate. Try reviewing [1], [2] to better understand the concept of creating a neural network policy for reinforcement learning.
- **Q and Deep Q Learning:** In Question 10, your response did not fully capture the primary challenges when adapting Q-value iteration to reinforcement learning scenarios with initially unknown transition probabilities and rewards. Try reviewing [3], [4] to gain a deeper understanding of the challenges of Q-learning and the solutions provided by deep Q networks.

Good job on the rest of the assignment!

Retry assignment

Instructions

1. Which of the following techniques are commonly used to prevent overfitting during the training of neural networks? Select all that apply. 3 / 3 point

☒ Dropout

☒ Correct

Correct: Dropout is a widely used regularization technique that helps prevent overfitting by reducing the model's reliance on specific neurons, encouraging more generalizable features.

☐ Lower learning rate

☒ L2 regularization (also known as weight decay)

☒ Correct

Correct: L2 regularization helps prevent overfitting by penalizing larger weights, which encourages simpler models that generalize better to unseen data.

☒ Early stopping

☒ Correct

Correct: Early stopping is a very effective method to prevent overfitting by halting training when the model's performance starts to degrade on the validation set, ensuring the model doesn't memorize the training data.

☐ Data normalization

2. What is the primary advantage of using Long Short-Term Memory (LSTM) networks over traditional Recurrent Neural Networks (RNNs) for sequence learning tasks? 3 / 3 point

☐ Reduced computational complexity

☐ Better for time-series forecasting

☒ Better at capturing long-range dependencies

☐ Increased internal memory

☒ Correct

Correct: LSTMs are specifically designed to mitigate the vanishing gradient problem, which enables them to capture and retain information over long sequences, an essential feature for tasks such as language modeling and time-series prediction.

3. Select all the benefits of using dropout in neural networks. 3 / 3 point

☒ Reduces overfitting by forcing the model to learn more generalizable features

☒ Correct

Correct: Dropout helps reduce overfitting by preventing neurons from co-adapting, which encourages the model to generalize better on unseen data.

☒ Promotes robust feature learning

☒ Correct

Correct: Dropout promotes more robust learning by preventing neurons from co-adapting, which leads to more diverse and effective feature representations.

☐ Reduces training time

☐ Increases the depth of the network

☒ Acts as an ensemble of networks

☒ Correct

Correct: Dropout effectively simulates training an ensemble of networks by using different subsets of the neural network during training, which helps improve the model's robustness.

4. What is the key reason for selecting actions randomly based on their probabilities in a neural network policy? 3 / 3 point

☐ It ensures the agent always chooses the most probable action.

☒ It allows the agent to explore new actions while still exploiting known good actions.

☐ It guarantees a 100% probability of selecting the best action.

☐ It minimizes the reward received by the agent.

☒ Incorrect

Correct: This approach helps the agent find a balance between trying new actions and sticking to actions that are already known to work well.

5. What is a key characteristic of a Markov process in reinforcement learning? 3 / 3 point

☐ It remembers all previous states to make decisions.

☒ It relies on a fixed transition probability between states, independent of past states.

☐ It evolves deterministically from one state to another.

☐ It evaluates policies by considering future rewards with no discount factor.

☒ Correct

Correct: Markov processes evolve based on fixed probabilities, depending only on the current state and action.

6. In the context of Markov processes, which term describes the property that the future state of the system depends solely on its current state, and not on the sequence of events that preceded it? 3 / 3 point

☒ The Markov property

☐ Temporal coherence

☐ State independence

☐ Sequential causality

☒ Correct

Correct: The Markov property refers to the memoryless nature of a Markov process, where the future state depends only on the present state and not on the past history of states.

7. Where does reinforcement learning fit in relation to supervised and unsupervised learning? 3 / 3 point

☐ It is a subset of supervised learning.

☐ It is entirely separate from both supervised and unsupervised learning.

☐ It is a hybrid of supervised and unsupervised learning.

☒ It is often considered a separate branch that lies between supervised and unsupervised learning.

☒ Correct

Correct: Reinforcement learning is often viewed as an independent branch that shares some characteristics with both supervised and unsupervised learning.

8. What is a major challenge faced by Q-learning when applied to large-scale Markov Decision Processes (MDPs)? 3 / 3 point

☒ High computational complexity

☐ Lack of convergence

☐ Inability to handle continuous action spaces

☐ Difficulty in reward design

☒ Correct

Correct: As the size of the state and action spaces in a large MDP grows, the computational complexity of Q-learning increases exponentially, particularly due to the need to maintain and update Q-values for every state-action pair, making it difficult to scale to large environments.

9. Which application of the Bellman equations is essential for estimating the optimal state and action values in the context of reinforcement learning? 3 / 3 point

☒ Evaluating long-term cumulative rewards

☐ Minimizing immediate costs

☐ Maximizing short-term rewards

☐ Balancing exploration and exploitation

☒ Correct

Correct: The Bellman equations provide a recursive framework for evaluating the expected long-term cumulative rewards of states and actions, considering both immediate and future rewards, which is central to finding the optimal policy in reinforcement learning.

10. What are the primary challenges when adapting Q-value iteration to reinforcement learning scenarios with initially unknown transition probabilities and rewards? Select all that apply. 0.8 / 3 point

☒ Exploration-exploitation trade-off

☒ Correct

Correct: In situations with unknown transition probabilities and rewards, an agent must strike a balance between exploring new state-action pairs to gather information and exploiting known strategies to maximize rewards. This trade-off is a fundamental challenge in reinforcement learning, particularly when Q-values are being iteratively updated.

☒ Computational efficiency

☒ Correct

Correct: As the agent explores the environment, the need to process and store Q-values for numerous state-action pairs can be computationally expensive. This challenge is exacerbated when the environment's dynamics (transition probabilities and rewards) are initially unknown, requiring additional learning and updating over time.

☐ Immediate reward maximization

☐ Accurate state representation

☐ Learning rate decay

You didn't select all the correct answers.